SIKORSKY UH-60M BLACKHAWK FOR X-PLANE

CDU / FMS



VERSION 0.9.12

THIS MANUAL EXPLAINS THE FUNCTIONS OF THE CDU/FMS USED IN THE "UH60M BLACKHAWK FOR X-PLANE". FOLLOWING REFERRED TO AS "UH60M-FMS".

THE UH60M-FMS IS A SIMPLIFIED VERSION OF THE VERY COMPLEX FMS USED IN THE REAL BLACKHAWK. IT IS NOT TRYING TO COPY THE REAL ONE, BUT IS PROVIDING SOME BASIC FUNCTIONS WHICH ARE SUPPORTED BY X-PLANE. THESE ARE: CONFIGURING AND USING COM1 AND COM2, NAV1 AND NAV2, XPDR AND RUNNING A FLIGHT PLAN.

IN THE UH60M-FMS THE FLIGHT PLAN IS SCRIPTABLE, WHICH MEANS BESIDE A STANDARD LIST OF WAYPOINTS, ADDITIONAL DEPENDENCY OF DATAREFS OR MODIFICATION OF DATAREFS OR CALLING COMMANDS CAN BE INCLUDED AS WELL. THESE FLIGHT PLANS (OR MISSIONS) CAN BE STORED IN THE "MISSIONS" FOLDER.

A "PRESETS.FPN" CAN BE USED TO DEFINE FREQUENCIES AND LOCATIONS BY NAME, SO THEY CAN BE LOADED MORE EASILY. THEIR NAMES ARE ALSO SHOWN IN THE UPPER "HEADER" PART OF THE FMS SCREEN.

AS MENTIONED BEFORE, THE SCREEN IS SPLIT INTO AN UPPER "HEADER" PART AND A LOWER DISPLAY PART WHICH SUPPORTS THE SOFT-KEYS.

THE HEADER PART CAN SHOW COM OR NAV SETTINGS FOR THE PILOT AND THE CO-PILOT DEPENDING ON WHICH PAGE HAS BEEN SELECTED.

THE COM PAGE ALLOWS TO DISPLAY AND CHANGE THE FREQUENCIES OF THE COMS AND THE XPDR. FOR THE XPDR ALSO THE MODE CAN BE SELECTED:



HSI source is GPS/FMS

The pilot is using preset #3 named HTM on COM1

THE NAV PAGE SHOWS INFOS ABOUT BEARING, WIND, ALTITUDE, GROUNDSPEED AND NEXT WAYPOINT.



WHEN SETTING COM, NAV OR ADF FREQUENCIES, ONE CAN SET THE FREQUENCY, THE PRESET NUMBER OR CHOOSE THE FREQUENCY BY NAME FROM THE PRESET LIST

THE FPN PAGE ALLOWS TO SHOW ACTIVE WAYPOINTS, EDIT WAYPOINTS (ADD, CHANGE, DELETE), SAVE AND LOAD FLIGHT PLANS AND RUN/HOLD FLIGHT PLAN PROCESSING.

Use BRT and DIM to control instrument brightness



IF A FLIGHT PLAN IS RUNNING, ITS NAME WILL BE SHOWN WITH INVERTED COLOR BELOW THE HEADER LINE:



THE LEFT SOFT-KEY COLUMN CAN BE USED TO SELECT WHICH WAYPOINT SHOULD BE SET AS NEXT TARGET. SELECTED WPT ALSO SHOWS THE REMAINING DISTANCE TO IT.

WHEN EDITING A FLIGHT PLAN YOU CAN ADD (APPEND TO LIST), INSERT BETWEEN 2 EXISTING WAYPOINTS OR DELETE A WAYPOINT FROM THE FLIGHT PLAN. ALT ALLOWS YOU ADD ALTITUDE AND VERTICAL SPEED, WHICH THE FLIGHT COMPUTER WILL USE TO GET TO THE WAYPOINT, IF THE FLIGHT PLAN IS RUNNING.



IF “AUTO_VS” IN ACTIVE, THE FMS WILL ADJUST VS DYNAMICALLY AS REQUIRED. OTHERWISE THE MESSAGE “UNABLE NEXT ALT” WILL APPEAR ON THE MFD IF VS IS NOT SUFFICIENT TO REACH THE NEXT TARGET AT THE SPECIFIED ALTITUDE.

THE CLC PAGE SHOWS CALCULATIONS ABOUT ETA AND FUEL USAGE OF THE CURRENT FLIGHT.

BESIDE USING THE ACTUAL KIAS, IT IS POSSIBLE TO CHOOSE DIFFERENT SPEEDS TO SEE HOW THE FUEL CONSUMPTION OR ETA WOULD CHANGE.



THE ZERO PAGE ALLOWS TO WIPE INTERNAL MEMORY OF THE CDU/FMS. IT CAN BE SELECTIVE LIKE MISSION OR FLIGHT PLAN ONLY (MSN) OR ALL.



THE GPS PAGE PROVIDES A LIST OF AIRPORTS NEARBY.
IT'S SORTED BY DISTANCE, SO CLOSEST FIRST.

Use PRV and NXT to scroll pages up and down



THE DRCT BUTTON ALLOWS YOU TO ENTER AN AIRPORT ID DIRECTLY.

IF A FLIGHT PLAN IS LOADED, THE MAP FUNCTION WILL SHOW A MAP OF THE WAYPOINTS. DOTTED LINES WE HAVE BEHIND US, SOLID LINES ARE IN FRONT OF US.
TRK SHOWS THE ACTUALLY FLOWN PATH IN A SOLID YELLOW LINE, LIKE ADS-B TRACKING.



ZOOMING IN AND OUT AND ALIGNING THE MAP TO NORTH OR HEADING IS SUPPORTED TOO.

THE DAT PAGE ALLOWS TO SHOW THE PRESETS:



COM PRESETS



NAV PRESETS



GPS PRESETS



THE DEFAULT PRESETS.FPN LOOKS LIKE THIS:

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10,1,TAC-OPS,123.220
10,2,KNIGHT-2,121.873
10,3,HTM,129.950
10,4,EDDM-ATIS,123.125
10,5,LOWS-ATIS,133.330
10,6,LOWI-ATIS,126.025
10,7,CTESTI,123.456
10,9,BANTAN9,124.60
10,15,KIL06,123.60
20,2,THE-BALL,108.10
20,3,TACAN-MAH,99X
20,4,VOR-MAH,115.20
20,5,VOR-MUN,112.30
20,6,ILS-ETSI,108.10
20,7,ILS-LOWI,111.10
20,8,ADF-MNW,338
20,9,ADF-MSW,400
30,11,MEGA8,48.1854,11.8631
30,12,KONUS,48.1789,11.9277
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VALID COMMAND TYPES:

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 0 MISSION-ID
 1 MISSION-CODENAME
 2 MISSION-DESCRIPTION
10 COM DEFINE PRESET
11 COM1 PRESET OR FREQ
12 COM2 PRESET OR FREQ
20 NAV DEFINE PRESET
21 NAV1 PRESET OR FREQ
22 NAV2 PRESET OR FREQ
23 ADF1 PRESET OR FREQ
24 ADF2 PRESET OR FREQ
30 LOCATION DEFINE PRESET
31 LAT/LON LOCATION FROM PRESET OR DIRECT
32 NAVAID LOCATION
33 DISCONTINUITY
40 MESSAGE
50 EXEC COMMAND, BELOW ALTLO OR ABOVE ALTHI
51 FIX COMMAND: DROPDUDE
60 SET DATAREF,IDX,VALUE,ALTLO,ALTHI
61 GET DATAREF,IDX,VALLO,VALHI
81 RESET MISSION
88 RESTART/REPEAT
99 NOP
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THE UH60M-FMS CAN READ X-PLANE ".FMS" FILES AND IT'S OWN FORMATS FOR FLIGHT-PLANS AND MISSIONS: ".FPN" AND ".MSN". SAVING IS ALWAYS DONE IN THE .FPN FORMAT.

THE FORMATS OF .FPN AND .MSN FILES ARE THE SAME. THE ONLY DIFFERENCE IS, THAT .MSN FILES (MISSIONS) START AUTOMATICALLY AT THE FIRST ENTRY (I.E. THE CHECKLIST.MSN, WHICH IS JUST A MISSION), WHILE A FLIGHT PLAN WILL WAIT UNTIL THE PILOT SELECTED THE WAYPOINT TO START WITH.

NOTE: IN THE FMS/CDU THE WPT PAGE ONLY SHOWS WAYPOINTS. ANY OTHER "COMMANDS" ARE HIDDEN AND CAN ONLY BE ADDED/CHANGED/DELETED BY EDITING THE FILE ITSELF.

MORE ABOUT MISSIONS AND THE FILE FORMATS CAN BE FOUND IN THE "MISSIONS" FOLDER IN THE FILE "MISSIONS_HOWTO.TXT".